GEK-83786C

DRIVE SYSTEM INSTRUCTIONS

POWER CIRCUIT BREAKER

DS2812B307
375VDC-900ADC

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

GEK-83786C

ISSUED: January 1983
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REVISED: November 1985

NOTICE

The information herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Department, 1501 Roanoke Boulevard, Salem, Virginia, USA, 24153.

NOTICE

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SAFETY SYMBOL LEGEND

********** * WARNING * **********	Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in personal injury or death.
********** * CAUTION * **********	Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in damage to, or destruction of equipment.
NOTE	Commands attention to an essential operating or maintenance procedure, condition or statement which must be highlighted.

INTRODUCTION

The breaker is used to apply and remove D.C. power from an exciter or generator field. The breaker power circuit consists of two (2) normally-open (N.O.) and one (1) normally-closed (N.C.) contacts. The contacts are in a make-before-break condiguration. There are 11 convertible auxiliary contacts available. The two (2) N.O. contacts apply D.C. power to the field. The N.C. contact inserts a discharge resistor into the field circuit. The breaker is mechanically latched when closed. There is an anti-pump circuit which will insure that there is only one closure per close command. The breaker is designed for easy access for maintenance purposes. The breaker is a non-drawout type, with a padlock feature to lock the breaker in the open position. With the breaker latched in, there are means for a mechanical as well as an electrical trip. Other relays have been incorporated which provide the anti-pump feature as well as provide auxiliary contacts for customer use. The breaker consists of two normally open IC2800EA900 contactors mechanically joined together which, when closed, opens the contacts of a normally closed DS303A5 contactor. Instruction books GEK-83756 and GEH-4286 should be used in conjunction with this instruction book.

****
* WARNING *
****

THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN. ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGHLY FAMILIAR WITH THE EQUIPMENT AND THESE INSTRUCTIONS SHOULD INSTALL, OPERATE, OR MAINTAIN THIS EQUIPMENT.

****
* WARNING *
****

When using this device for replacement, care should be taken to insure power circuits are connected as shown in Figure 3.

RECEIVING AND HANDLING

Immediately upon receipt, the equipment should be carefully unpacked and examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric Sales Office should be notified promptly. If the breaker is to be stored, it should be protected from debris, in a dry area.

INSTALLATION

The breaker should be carefully inspected before installation to see that parts and adjustments, as indicated in these instructions, are in proper operating condition. Any packing material or other foreign matter should be removed, particularly from the contact tip and armature assemblies.

Grease or other foreign material must be removed from the armature and magnet mating surfaces. Use a cloth dampened with a suitable solvent such as xylol. The armature and magnet should then operate positively and quietly in the closed position. The armature should be closed manually to be sure tht the contactor operates without excessive friction or binding.

The breaker must be mounted vertically. Care should be taken to insure that the breaker is protected from debris which may jam contacts.

RATING

Maximum continuous D.C. current rating - N.O. poles . . . .
900A

Maximum continuous D.C. current rating - N.C. pole. . . . .
300A

Maximum continuous D.C. voltage rating. . . . . .
375VDC

Maximum interrupting current--also current closing
contact must carry for 5 seconds. . . . . .
3000A

Maximum interrupting voltage. . . . . .
1500V

Close time (seconds). . . . . .
.200

Trip time (seconds) . . . . . .
.033

D.C. Control Power. . . . . .
125/250VDC

DESCRIPTION

The breaker consists of two major circuits, a trip circuit, and a close circuit. There is an anti-pump circuit and a circuit to provide auxiliary contacts, which work in conjunction with the close and trip circuit.

For the following discussion of operation, Figure 1 may be used.

Apply D.C. control power to the breaker. Place the 41 switch in the close position. Relay 41X will energize through the normally closed contacts of 41Y, 8PD and 41 Auxiliary. 41X energizing will close its contacts in series with the CLOSE coils 41CCA and 41CCB. In order for the breaker to mechanically latch, 41CC must remain energized for approximately 250 milliseconds. This is accomplished by 41X being a time delay relay. Relay 41Y will insure only one closure of the main poles per CLOSE command. The above sequence may be performed by operating the CLOSE pushbutton located on the face of the breaker. Once the breaker has picked up, it is mechanically latched in. If the D.C. control power were removed, the breaker will remain latched in. With power removed, the breaker may be tripped by operation of the TRIP pushbutton on the face of the breaker.

With DC power applied, place the 41 switch in the TRIP position. The 41 Auxiliary contacts 1-1C, 2-2C will be closed due to the breaker being in the closed position. Closing the 41 switch will allow 41TC to be energized. This action will open the breaker. As stated before, the breaker may be opened by operation of the TRIP pushbutton on the face of the breaker.

Relay 41R1 will follow the action of the breaker with one exception. If the DC power is removed and the breaker is mechanically tripped, 41R1 will not change state.

Points are wired out for connection of indicating lights. The contact labelled 8PD is mechanically opened when the breaker is padlocked open. This prevents the breaker from being accidentally energized.

MAINTENANCE/ADJUSTMENTS

The breaker will provide maximum service if inspected and maintained on a regular basis. The life of the breaker depends upon the severity of its service. Instruction books GEK-83756 and GEH-4286 should be used in conjunction with the following adjustment procedure.

1. Main contactor Alignment
   A. De-energize both contactors and check that when one armature is resting against its stop, the other armature has a .030 inch maximum gap with its stop. Shim the stop as required.
   B. Verify freedom of operation by hand closing armatures. There should be a minimum side-to-side clearance of .010 inch.

2. With the contactor picked up and "held in" electrically, adjust plunger "B" to obtain a minimum gap at "A" (.01-.02 inch). At no time is the armature to bottom against the stop. Be sure to lock plunger (B) in place with nut (C) after adjusting.

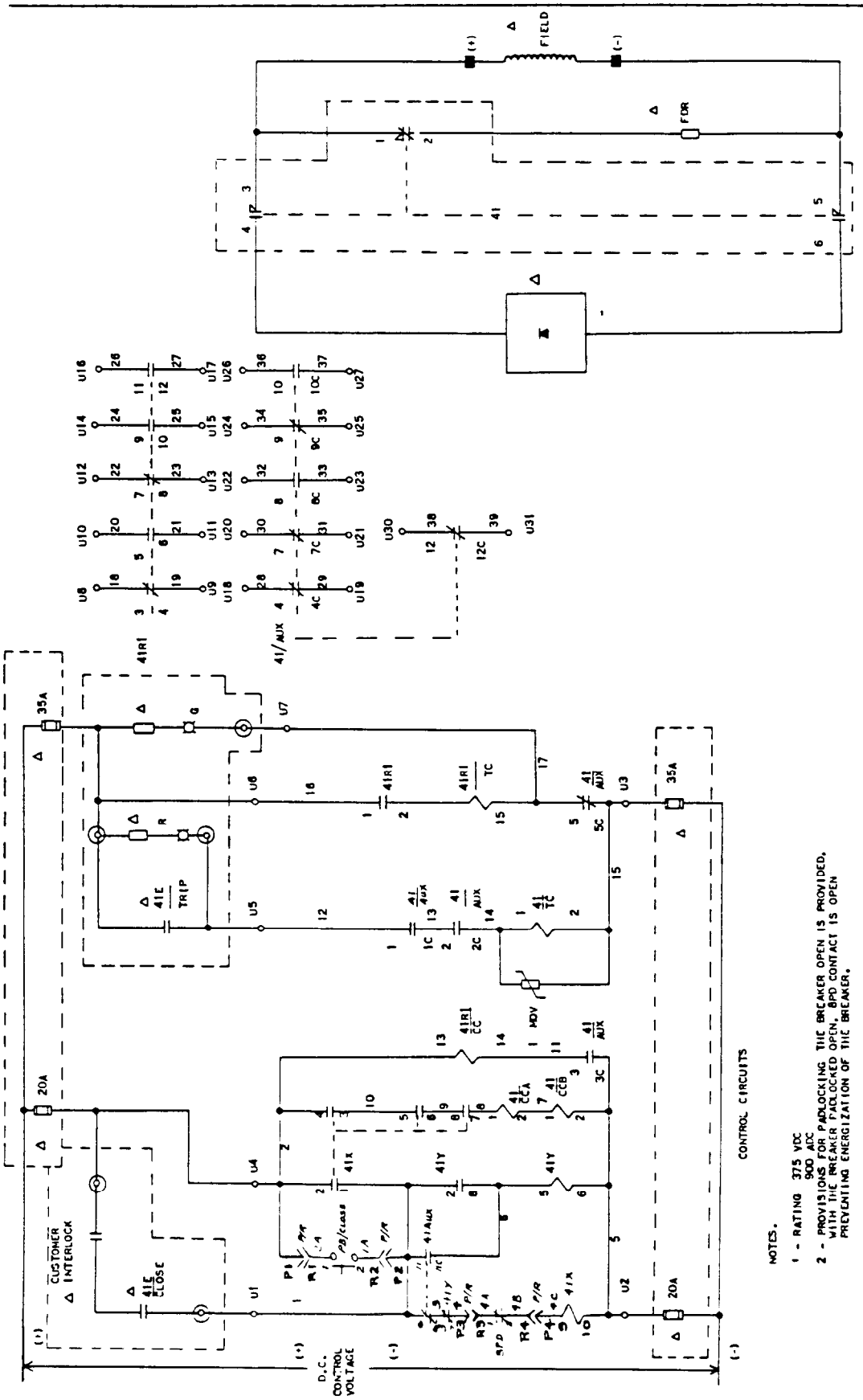

FIELD
(+)
(-)
FDR
41/AUX
41R1
CONTROL CIRCUITS
CUSTOMER INTERLOCK
41E CLOSE
41E TRIP
D.C. CONTROL VOLTAGE
NOTES.
1 - RATING 375 VDC
900 ADC
2 - PROVISIONS FOR PADLOCKING THE BREAKER OPEN IS PROVIDED, WITH THE BREAKER PADLOCKED OPEN, &PD CONTACT IS OPEN PREVENTING ENERGIZATION OF THE BREAKER.
Δ - NOT PROVIDED BY DSD

Figure 1

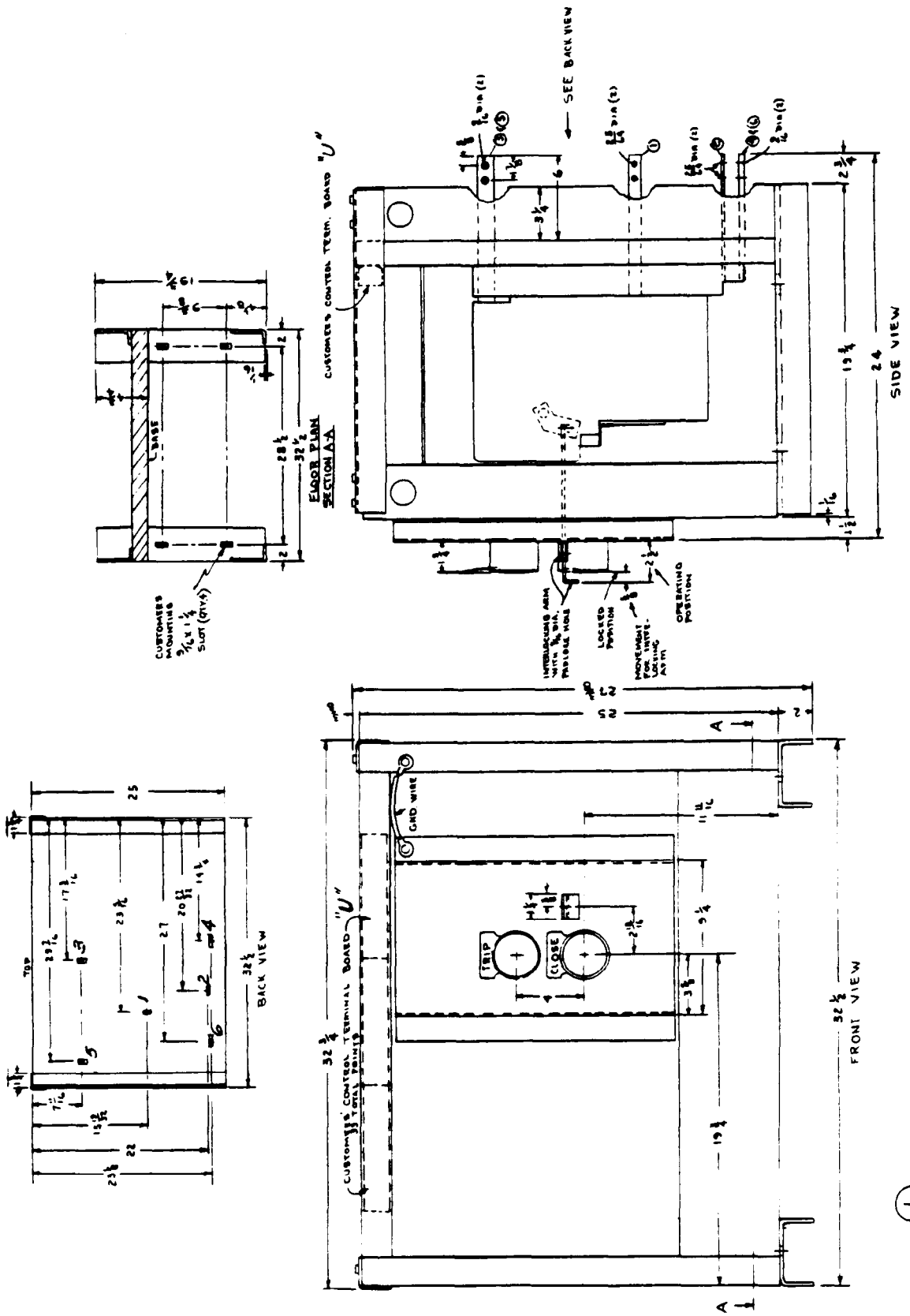


Figure 2
900A Field Contactor Outline Drawing

3. With the contactor still picked up electrically, set gap at "D" to .05 to .08 inch by placing shims between roller arm and bracket as shown in Figure 3.

   Extra shims are provided for field adjustment.

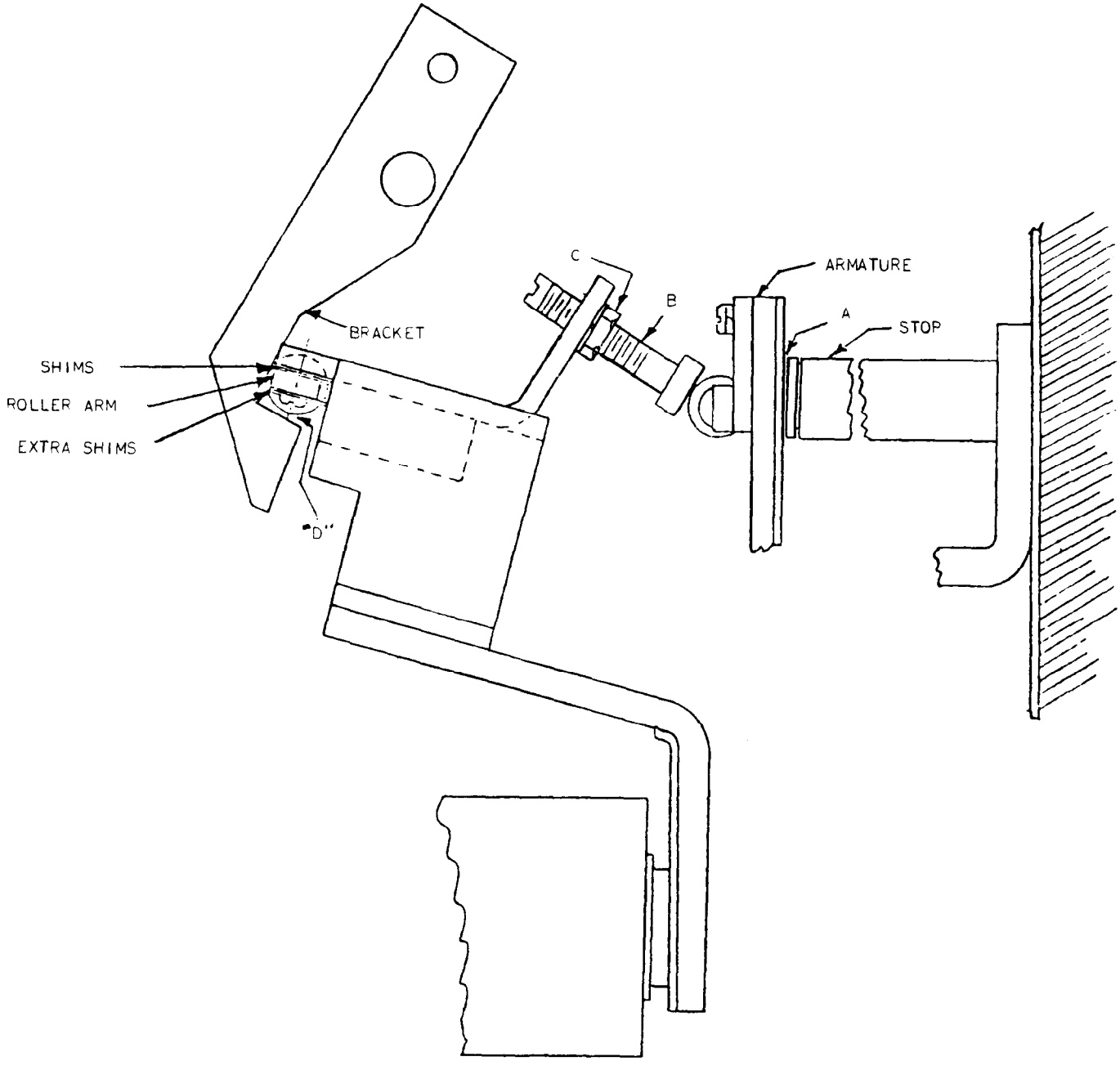


Figure 3

4. With the contactor mechanically latched in place and coil voltage removed, minimum gap "A" should be at least .42 inch measured at its narrowest point. If the gap is less than .42, adjust plunger "B" as in paragraph 2 to obtain necessary gap (See Figure 4).

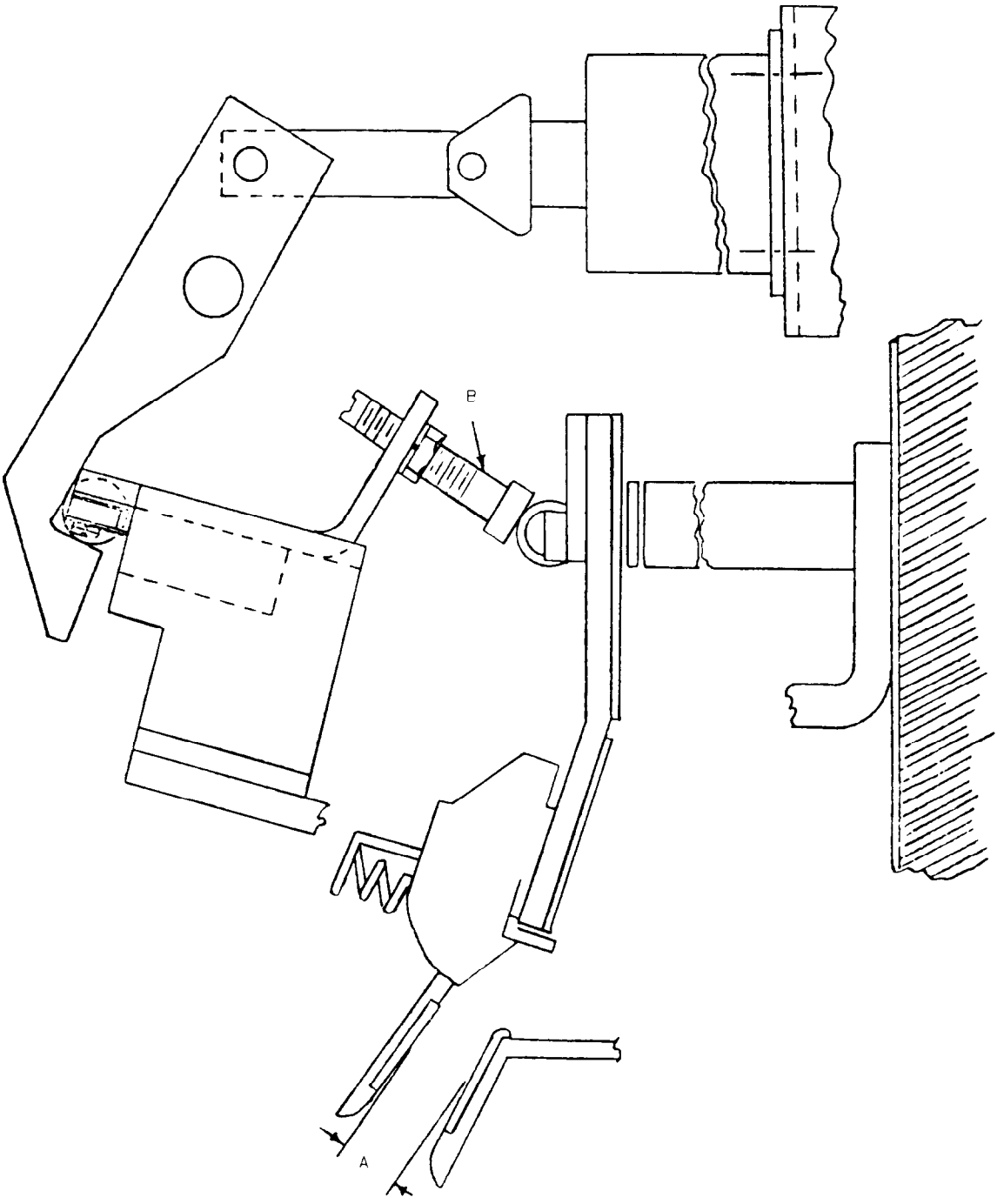


Figure 4

5. With the contactor in the unlatched position, depress (bottom) the solenoid plunger to create the maximum gap between the latch arm and the roller. Shim solenoid mounting as needed to obtain .18 inch ±.06 inch gap at roller (See Figure 5).

Set gap at (G) to be .051 to .081 inches by adjusting the armature stop.

After the gap is set, adjust springs (2) to positively seat the armature against the armature stop.

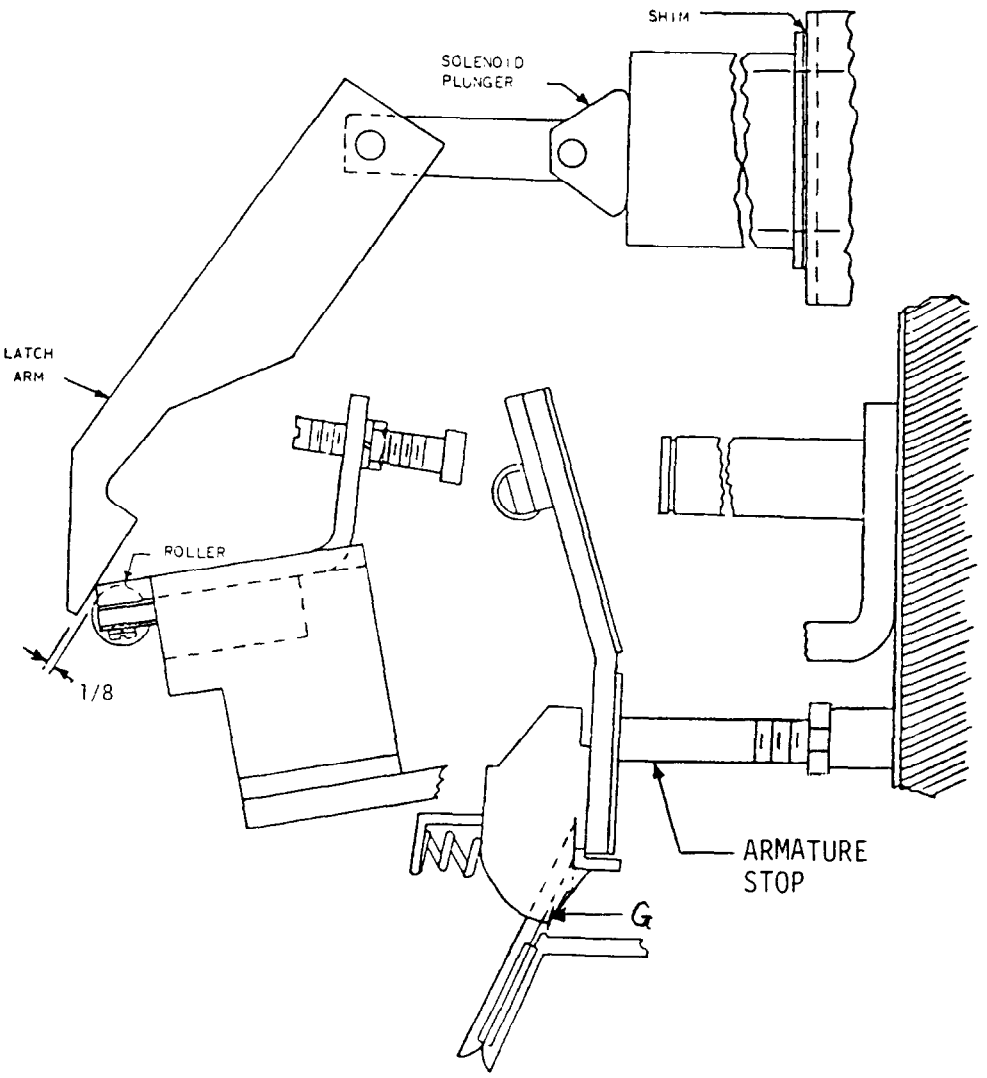


Figure 5

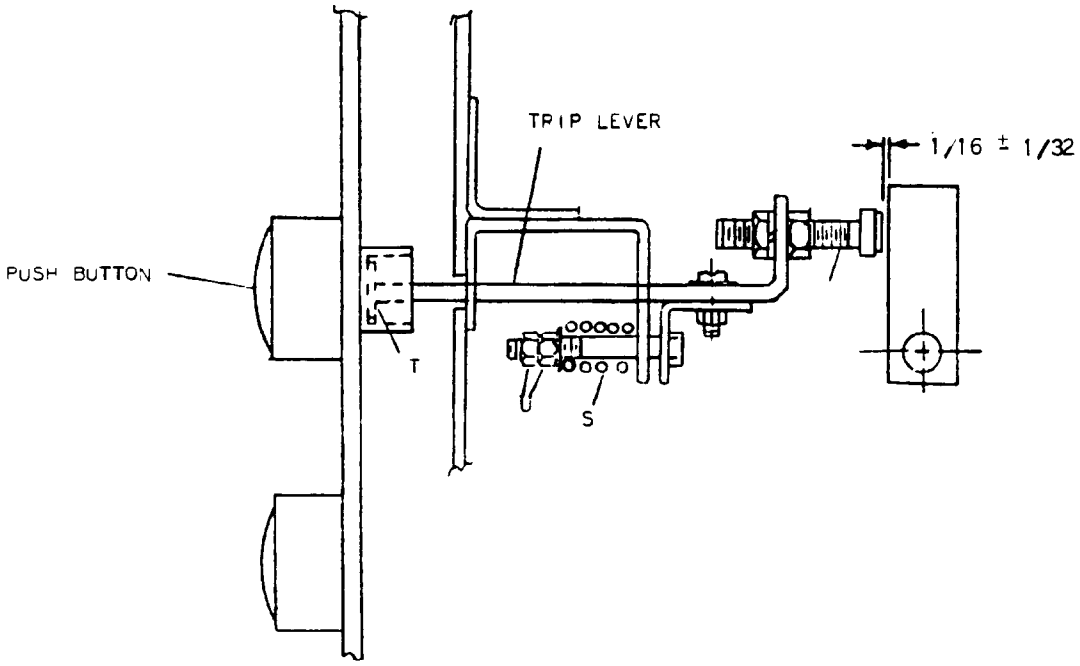


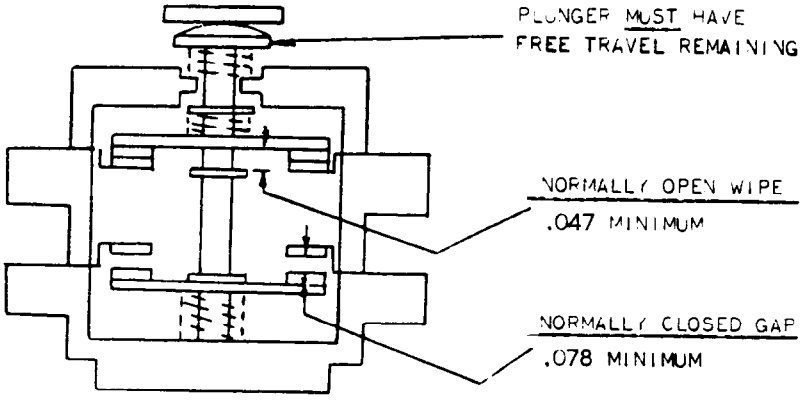
Figure 6

6. Adjust the bolt - nut assembly "U" to depress spring "S" in order to make sure trip lever is pressing against the pushbutton assembly at point "T".

7. With the contactor mechanically latched closed, adjust bolt "V" until a gap of 1/16 $\pm$ 1/32 inch is obtained. Verify that the contactor can be unlatched by depressing the push button.

8. With the contactors picked up electrically, the auxiliary interlock plungers must have free travel remaining.

   With the contactor mechanically latched closed, verify minimum auxiliary interlock gap and wipe.



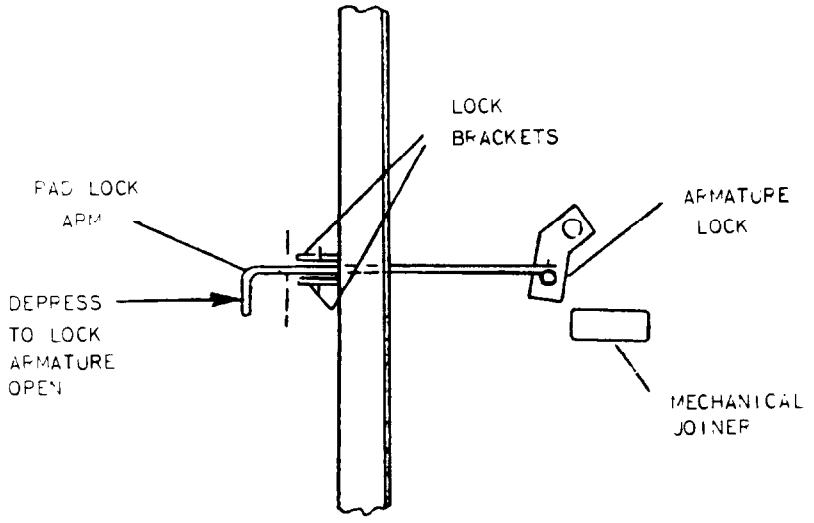


Figure 7

9. Padlock function Check:

   Depress padlock arm to make sure the armature lock rotates above the mechanical joiner. Check that the hole in the padlock arm aligns with the holes in the lock brackets, and that positive armature lock engagement is attained with a .06 pin in the padlock hole.

   Release the padlock arm. The armature lock should clear the path of mechanical joiner when the joiner assembly is rotated. Repeat depressing and releasing operation several times to assure that no binding occurs in the area of the lock or in the area of the lock brackets.

With D.C. power applied, place the 41 switch in the TRIP position. The 41 Aux 2 and 2C contact will be closed due to the breaker being in the closed position. Closing the 41 switch will allow 41TC to be energized. This action will open the breaker. As stated before, the breaker may be opened by operation of the TRIP pushbutton on the face of the breaker.

Relay 41R1 will follow the action of the breaker with one exception. If the D.C. power is removed and the breaker is mechanically tripped, 412R1 will not change state.

Connection points are wired out for connection of indicating lights. The contact labelled 8PD is mechanically opened when the breaker is padlocked open. This prevents the breaker from being accidentally energized.

DS2812B307 POWER CIRCUIT BREAKER
RECOMMENDED SPARE PARTS

QUANTITY	CATALOG NUMBER	DESCRIPTION
2	117A7313G1	Arc Chute Assembly
2	259A5562G1	Stationary Contact
2	273A2426G1	Movable Contact
2	171B3805G2	Shunt
2	259A5572P1	Spring for Contact
2	2412649	Return Spring
2	22D75G15A 22D75G7A	Coil. 125V Control Coil, 250V Control
1	195B6285G2	Arc Chute Assembly
1	259A5658G1	Stationary Contact
1	259A5663G1	Movable Contact
1	304A6610G1	Shunt
1	259A5676P1	Spring for Contact
1	CR9500B202U3A CR9500B202U5A	Solenoid, 125V Control Solenoid, 250V Control
1	12HGA11J52 12HGA11J51	HGA Relay, 125V Control HGA Relay, 250V Control
1	IC2820A100EM203L IC2820A100EM202L	Time Delay Relay, 125V Control Time Delay Relay, 250V Control
8	DS2956A205C	1NO, 1NC Universal Interlock
1	12HFA54E187H 12HEA54F218H	HFA Relay, 125V Control HFA Relay, 250V Control

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